Distinct distances in general position: an explicit improvement of the bound $n/3$

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October 2026

Abstract

Place n points in the plane so that no three are on a line and no four are on a circle. How many different distances must they determine? Every circle centred at one of the points can pass through at most three of the others, so the $n-1$ other points need at least $(n-1)/3$ circles, and there are at least $(n-1)/3$ distinct distances. I prove that every such set with $n \geq 10^4$ points determines more than $(1/3+1/40)\,n$ distinct distances. The proof first shows that a set with few distances must lie in a thin ring (an annulus) around its centre of mass. It then derives many linear inequalities that the distribution of the pairwise distances must satisfy, and shows that no distribution satisfies all of them. The last step is an explicit certificate: nonnegative rational weights such that the weighted sum of the inequalities is false for every possible profile. The certificate has been checked by an independent exact verifier, and the complete proof, with the certificate, has been checked by the Lean 4 proof assistant.

1 Introduction

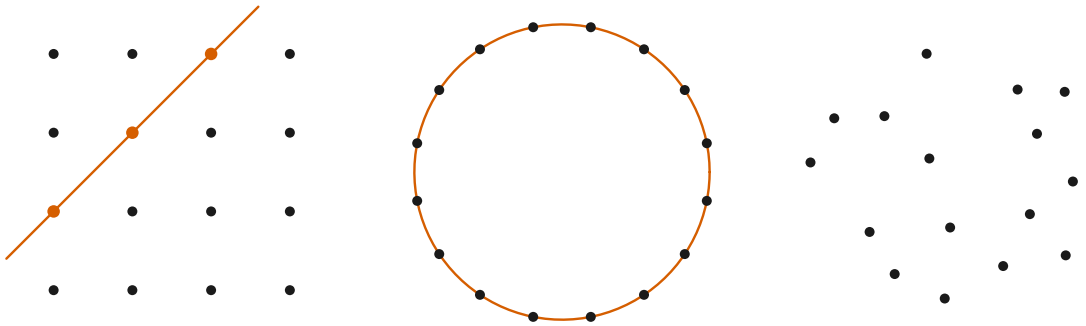
1.1 The question

For a finite set P of points in the plane, let $D(P)$ be the number of *distinct* distances between pairs of its points: we list the distance $|pq|$ for every pair $p \neq q$ and count how many different values occur. A set of n points has $\binom{n}{2}$ pairs, so $D(P) \leq \binom{n}{2}$, and for a typical set equality holds. The interesting question goes the other way: how *small* can $D(P)$ be? This is the distinct distances problem of Erdős.

The answer depends on what the set is allowed to look like. Figure 1 shows three sets of 16 points. The square grid repeats each distance many times and has only 9 distinct distances. The vertices of a regular polygon have only 8. A set with no special structure has $\binom{16}{2} = 120$. Both small examples rely on structure: the grid has many points on a common line, and the polygon has all its points on a common circle.

We say that P is *in general position* if no three points of P are collinear and no four points of P are concyclic (on one circle). Erdős asked whether every set of n points in general position determines a number of distinct distances that grows faster than linearly in n [2, question (9)]. This is Problem 98 on the Erdős problems website [6], where it is listed as open.

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(a) $D = 9$ (b) $D = 8$ (c) $D = 120$

Figure 1: Three sets of 16 points and their numbers of distinct distances. (a) The 4×4 grid has $D = 9$; it contains three points on a line (drawn). (b) The regular 16-gon has $D = 8$; all its points lie on one circle. (c) 16 points in general position: no three on a line, no four on a circle. Here every distance occurs once, so $D = 120$.

1.2 The easy bound $n/3$

Fix a point $p \in P$ and draw, for every distance r that occurs in P, the circle with centre p and radius r (Figure 2). Every other point q of P lies on exactly one of these circles, the one of radius $|pq|$. If four points of P were on one of these circles, they would be concyclic. So each circle holds at most three points of P, and the $n-1$ other points need at least $(n-1)/3$ circles:

$$D(P) \;\geq\; \frac{n-1}{3}.$$

Brass, Moser and Pach record this bound and state that no improvement of the form $(1/3+\varepsilon)n$ is known, both for sets with no four points on a circle and for sets in general position [1, §5.5]. For sets with no three points collinear, Erdős reports an argument of Szemerédi that gives a point with more than $\lfloor n/3 \rfloor$ distances from it [2]. I am not aware of a bound of the form $(1/3+\varepsilon)n$ for sets in general position. In the other direction, Brass, Moser and Pach list the upper bound $ne^{O(\sqrt{\log n})}$ for such sets, due to Erdős, Füredi, Pach and Ruzsa [4]. Without the general position hypothesis the minimum lies between $cn/\log n$, by the theorem of Guth and Katz [7], and $Cn/\sqrt{\log n}$, attained by the square lattice.

1.3 The result

Theorem 1. *Let $P \subset \mathbb{R}^2$ be a set of $n \geq 10^4$ points, no three collinear and no four concyclic. Then*

$$D(P) \;>\; \Big(\frac{1}{3} + \frac{1}{40}\Big)\, n.$$

The proof is by contradiction: it assumes a set of $n \geq 10^4$ points in general position with $D(P) \leq (1/3 + 1/40)n$ and derives a contradiction. The constant $1/40$ and the threshold 10^4 are explicit, and both enter through the certificate of Section 8.

1.4 How the proof works

This section explains the idea of the proof without formulas. Figure 3 shows its structure.

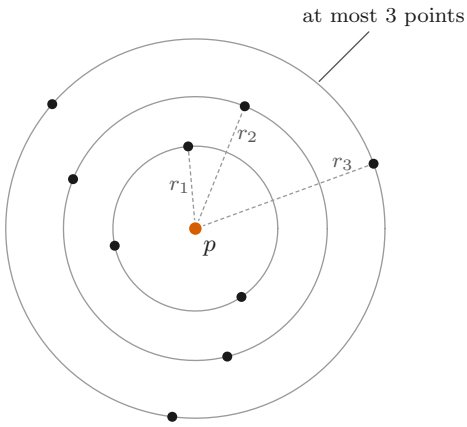


Figure 2: The easy bound. Around a point p, every other point lies on a circle centred at p whose radius is one of the distances of the set. In general position each such circle holds at most three points, so the $n-1$ other points need at least $(n-1)/3$ distances. In this example every drawn circle holds exactly three points, the most that general position allows.

Step 1: few distances leave few empty slots. Think of every distance r in the set as providing three "slots" on the circle of radius r around p. The $n-1$ other points fill $n-1$ of the $3D$ slots, so $Q = 3D - (n-1)$ slots around p stay empty. This number is the same for every point p. If D is at most $(1/3+\delta)n$, then $Q \leq 3\delta n + 1$: only a small fraction of the slots are empty. With $\delta = 1/40$, at every point at least about 79% of the D circles hold exactly three points. This is a rigid structure, and the rest of the proof shows that it cannot exist. One consequence (Section 2) concerns isosceles triangles: at least about 85% of the pairs of points $\{a, b\}$ are the base of exactly two isosceles triangles whose apex is a point of the set.

Step 2: the set lies in a thin ring. The sum of the squared distances from a point p to the other points measures how far p is from the rest of the set. A classical identity (the parallel axis theorem of mechanics) says that this sum equals $n|p|^2 + S$, where $|p|$ is the distance from p to the centroid (centre of mass) of the set and S does not depend on p. So points far from the centroid have larger sums. On the other hand, if all slots around p are full, the sum is $3\sum_r r^2$, summed over all the distances r of the set, which also does not depend on p. So if there were no empty slots at all, every point would be at the same distance from the centroid: the whole set would lie on one circle, which general position forbids. With few empty slots, the points are nearly at the same distance from the centroid: after scaling, they lie in the *annulus*

$$0.836 \;\leq\; |p| \;\leq\; 1,$$

a ring of width about 0.16 around the centroid. Section 3 proves this, and Figures 6 and 7 illustrate it. The annulus is the main tool of the proof: a circle of medium radius around a point of the ring meets the ring in two short arcs, and a line that passes near the centre meets the ring in two short segments. This gives many geometric restrictions.

Step 3: inequalities for the distance profile. The proof does not work with individual points. It works with the *distance profile*: for each ρ between 0 and 2, the fraction $\mathcal{A}(\rho)$ of pairs of points at distance at most ρ, and the number of distinct distances at most ρ. These two functions are like cumulative distribution functions. Each geometric fact about the ring gives an inequality between

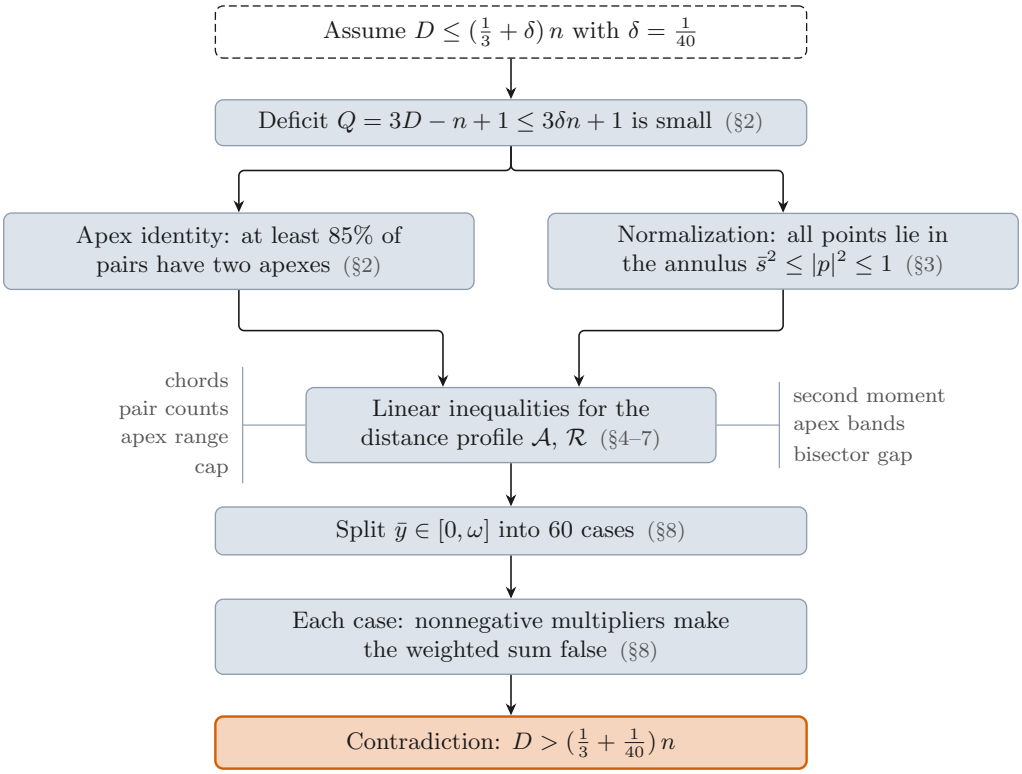


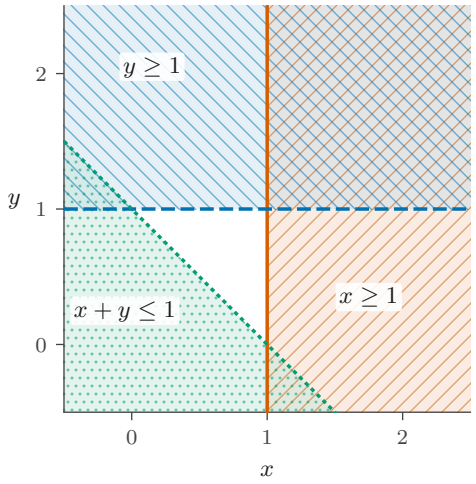
Figure 3: The structure of the proof. Each box names the section where the step is proved.

values of these functions. Some examples: a full circle of medium radius around a point of the ring must contain two close points (Section 4); a long pair must have its two apexes at distances in a known range (Section 5); and the two apexes of a long pair are either very close to each other or very far apart (Section 7). All these inequalities are linear in the unknown values of the profile.

Step 4: no profile satisfies all the inequalities. We now have a system of linear inequalities in finitely many unknowns, and we want to show that it has no solution. There is a simple way to prove that a system of linear inequalities has no solution: multiply each inequality by a nonnegative number and add them up. If the result is a false statement such as $0 \le -1$, the system has no solution, because any solution would also satisfy the sum. Figure 4 shows a small example. The list of multipliers is called a *certificate*. A computer finds the multipliers for our system, but the computer is not needed to trust the result: once the multipliers are written down, anyone can check them with exact arithmetic on fractions. Section 8 describes the certificate, and Section 9 describes two checks of it: an exact verifier that does not use the program that found the multipliers, and a formal proof in the Lean proof assistant.

Reading guide. Sections 2 and 3 need only plane geometry and sums. Sections 4–7 derive the inequalities; each lemma is preceded by a paragraph that says what it means. Section 8 assembles the certificate. Appendix A lists the notation.

Remark on weaker constants. With fewer inequalities the same method gives weaker constants without a certificate file: Lemma 13 at one radius, Lemma 8(b) with a uniform chord bound,



$-x \leq -1$	multiplier 1
$-y \leq -1$	multiplier 1
$x + y \leq 1$	multiplier 1
$0 \leq -1$	the sum

No point lies in all three half-planes.

Figure 4: A certificate in a small example. The three inequalities $x \geq 1$, $y \geq 1$ and $x + y \leq 1$ have no common solution: the three shaded half-planes have no common point. Written as $-x \leq -1$, $-y \leq -1$, $x + y \leq 1$ and added with multipliers $1, 1, 1$, they give $0 \leq -1$. The proof of Theorem 1 does the same with 8964 unknowns, in 60 cases; the certificate of each case combines between 1083 and 4193 inequalities.

Lemma 9 at $\lambda = \sqrt{3}$ and Lemma 11 combine in 15 explicit inequalities to $D(P) > (1/3 + 1/60)n$ for $n \geq 10^4$. That derivation is not reproduced here.

2 Empty slots and isosceles triangles

Throughout, $P \subset \mathbb{R}^2$ is a set of $n \geq 2$ points in general position, R is the set of distances determined by P, and $D = |R|$. For $p \in P$ and $r > 0$ let $C(p, r)$ be the circle with centre p and radius r, and

$$d_r(p) = |P \cap C(p, r)|, \qquad \mathrm{def}_r(p) = 3 - d_r(p).$$

So $d_r(p)$ is the number of points of P at distance exactly r from p, and $\mathrm{def}_r(p)$ is the number of empty slots on that circle. Four points of P on $C(p, r)$ would be concyclic, so $d_r(p) \leq 3$ and $\mathrm{def}_r(p) \geq 0$. Every $q \in P \setminus \{p\}$ lies on exactly one circle $C(p, r)$ with $r \in R$, so $\sum_{r \in R} d_r(p) = n - 1$ and

$$\sum_{r \in R} \mathrm{def}_r(p) \;=\; 3D - n + 1 \;=:\; Q \qquad \text{for every } p \in P. \tag{1}$$

In particular $Q \geq 0$, which is the bound $D \geq (n - 1)/3$. We call Q the *deficit*. It is the number of empty slots around each point, and it is the same for all points.

Isosceles triangles. For distinct $a, b \in P$ let

$$\alpha(a, b) = \#\{z \in P \setminus \{a, b\} : |za| = |zb|\},$$

the number of points z of P that are the apex of an isosceles triangle with base ab. Such apexes lie on the perpendicular bisector $\mathrm{bis}(a, b)$, which is a line, and no three points of P are collinear, so $\alpha(a, b) \leq 2$ (Figure 5(a)). Let $N = \binom{n}{2}$ be the number of unordered pairs and let x_k be the number of unordered pairs with $\alpha = k$, so $N = x_0 + x_1 + x_2$.

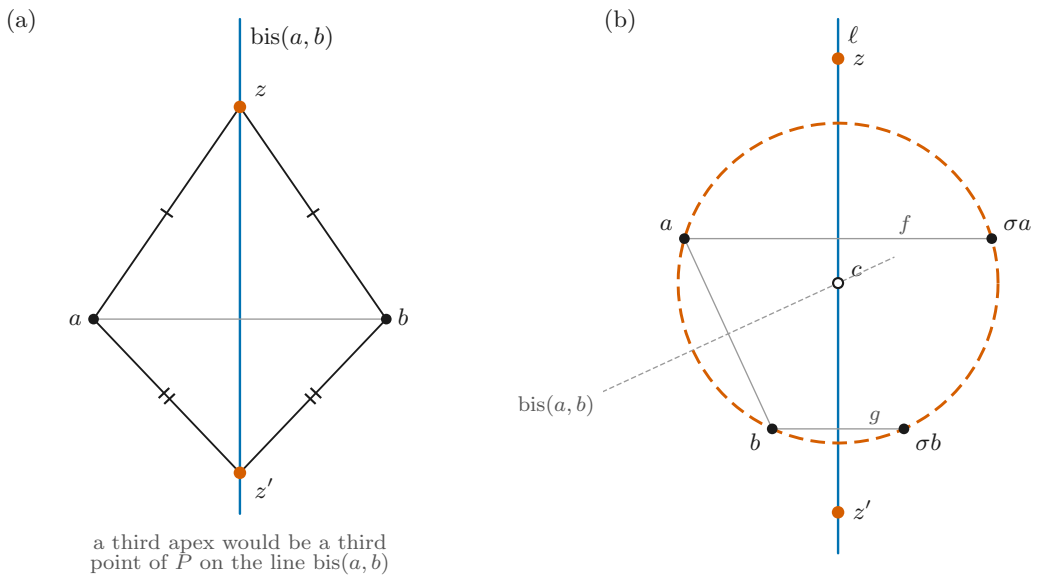


Figure 5: (a) The apexes z, z' of the isosceles triangles with base ab lie on the perpendicular bisector of ab. A third apex would put three points on one line, so a pair has at most two apexes. (b) Why two different pairs cannot have the same two apexes (Lemma 3). If the pairs $f = \{a, \sigma a\}$ and $g = \{b, \sigma b\}$ both have the apexes z, z', each pair is symmetric about the line $\ell = zz'$, and the point c where the bisector of ab meets ℓ is equidistant from all four points: four points on a circle.

What the next lemma says. The total number of "missing apexes", $\sum(2-\alpha)$ over all pairs, is at most n times the number Q of empty slots per point. When $D \le (1/3+\delta)n$, this says that at most about $3\delta n^2$ pairs have fewer than two apexes; with $\delta = 1/40$ that is at most about 15% of the $N \approx n^2/2$ pairs. The idea of the proof: a point z is an apex of the pair $\{a, b\}$ exactly when a and b lie on a common circle around z, and a full circle with three points supplies three such pairs.

Lemma 2 (apex identity)**.** *Let* $V = \sum_{p\in P}\sum_{r\in R} v(d_r(p))$ *with* $v(0) = 3$, $v(1) = 1$, $v(2) = v(3) = 0$. *Then*

$$\sum_{\{a,b\}} \big(2 - \alpha(a,b)\big) + V \;=\; nQ,$$

the sum running over the N *unordered pairs. Consequently* $N - x_2 \le nQ$, *and for every set* Φ *of ordered pairs of distinct points that is closed under* $(a,b) \mapsto (b,a)$,

$$\sum_{(a,b)\in\Phi} \big(2 - \alpha(a,b)\big) \;\le\; 2nQ. \tag{2}$$

Proof. Count the pairs (apex z, unordered pair $\{a,b\} \subset C(z,r)$): a circle $C(z,r)$ with d points gives $\binom{d}{2}$ pairs, so $\sum_{\{a,b\}} \alpha(a,b) = \sum_z \sum_{r\in R} \binom{d_r(z)}{2}$. Also $2N = \sum_z \sum_{r\in R} d_r(z)$, because each point z has $n-1$ partners. With (1),

$$nQ - \sum_{\{a,b\}} (2-\alpha) = \sum_z \sum_{r\in R} \Big[(3-d) - d + \tbinom{d}{2}\Big]_{d=d_r(z)},$$

and $3 - 2d + \binom{d}{2}$ equals $3, 1, 0, 0$ for $d = 0, 1, 2, 3$. For the consequences: $2 - \alpha \ge 0$ and $V \ge 0$, so $N - x_2 = x_0 + x_1 \le 2x_0 + x_1 = \sum(2-\alpha) \le nQ$; and the sum in (2) is at most the sum over all ordered pairs, which is $2\sum_{\{a,b\}}(2-\alpha) \le 2nQ$. □

The map κ. Let $\mathcal{D}$ be the set of unordered pairs f with $\alpha(f) = 2$, so $|\mathcal{D}| = x_2$. For $f \in \mathcal{D}$ let $\kappa(f)$ be the pair of points of P on $\mathrm{bis}(f)$, that is, the pair of its two apexes. The next lemma says that the two apexes determine the pair. It is used in Section 7.

Lemma 3. *κ is injective on $\mathcal{D}$.*

Proof. Let $f \neq g \in \mathcal{D}$ with $\kappa(f) = \kappa(g) = \{z, z'\}$, and let ℓ be the line zz' (Figure 5(b)). Then $\mathrm{bis}(f) = \mathrm{bis}(g) = \ell$, so with σ the reflection in ℓ we have $f = \{a, \sigma a\}$ and $g = \{b, \sigma b\}$. If f and g share a point, then $f = g$; so they are disjoint. If a and b lie on one line perpendicular to ℓ, the four points $a, \sigma a, b, \sigma b$ are collinear, which general position excludes. Otherwise $\mathrm{bis}(a, b)$ is not parallel to ℓ and meets ℓ at a point c, and $|ca| = |cb| = |c\sigma a| = |c\sigma b|$: four points of P on a circle, which general position also excludes. □

3 The annulus

This section proves Step 2 of the overview: a set with few distinct distances lies in a thin ring around its centroid. Everything here uses only the two expressions (3) and (4) below for one sum, and the count (1).

Translate P so that its centroid is the origin, $\sum_p p = 0$, and put $S = \sum_p |p|^2$. Translation does not change distances, collinearity or concyclicity.

Two ways to compute the same sum. For a point p, consider the sum $\sum_{q \in P} |p - q|^2$ of the squared distances from p to all points of P. On one hand, expanding $|p - q|^2 = |p|^2 - 2\langle p, q\rangle + |q|^2$ and using $\sum_q q = 0$,

$$\sum_{q \in P} |p - q|^2 \;=\; n|p|^2 + S. \tag{3}$$

(This is the parallel axis theorem: the moment of inertia about p is the moment about the centroid plus n times the squared distance from the centroid.) So the sum grows with the distance $|p|$ from p to the centroid. On the other hand, grouping the points q by their distance r from p,

$$\sum_{q \in P} |p - q|^2 \;=\; \sum_{r \in R} d_r(p)\, r^2 \;=\; 3\sum_{r \in R} r^2 \;-\; \sum_{r \in R} \mathrm{def}_r(p)\, r^2. \tag{4}$$

The first term on the right, $3\sum_r r^2$, is the same for every point p: it would be the value of the sum if every circle around p were full. The second term is the "missing moment" of the empty slots at p. If there were no empty slots, (3) and (4) would give the same $|p|$ for every p, so all points would lie on one circle around the centroid. If there are few empty slots, the missing moment is small, and all points are *nearly* at the same distance from the centroid. Figure 6 shows the effect in random examples. The lemmas below make this precise.

Lemma 4. *$\rho_A^2 := \big(3\sum_{r \in R} r^2 - S\big)/n$ is positive.*

Proof. Summing (3) over p gives $\sum_{p,q} |p-q|^2 = 2nS$. By (4) and $d_r(p) \le 3$, $\sum_q |p-q|^2 \le 3\sum_{r \in R} r^2$. Summing over p gives $2nS \le 3n\sum_r r^2$, so $3\sum_r r^2 - S \ge S$, and $S > 0$ because P has at least two points and centroid 0. □

Scale P by $1/\rho_A$. Scaling preserves general position and D, and after it

$$3\sum_{r \in R} r^2 = n + S. \tag{N1}$$

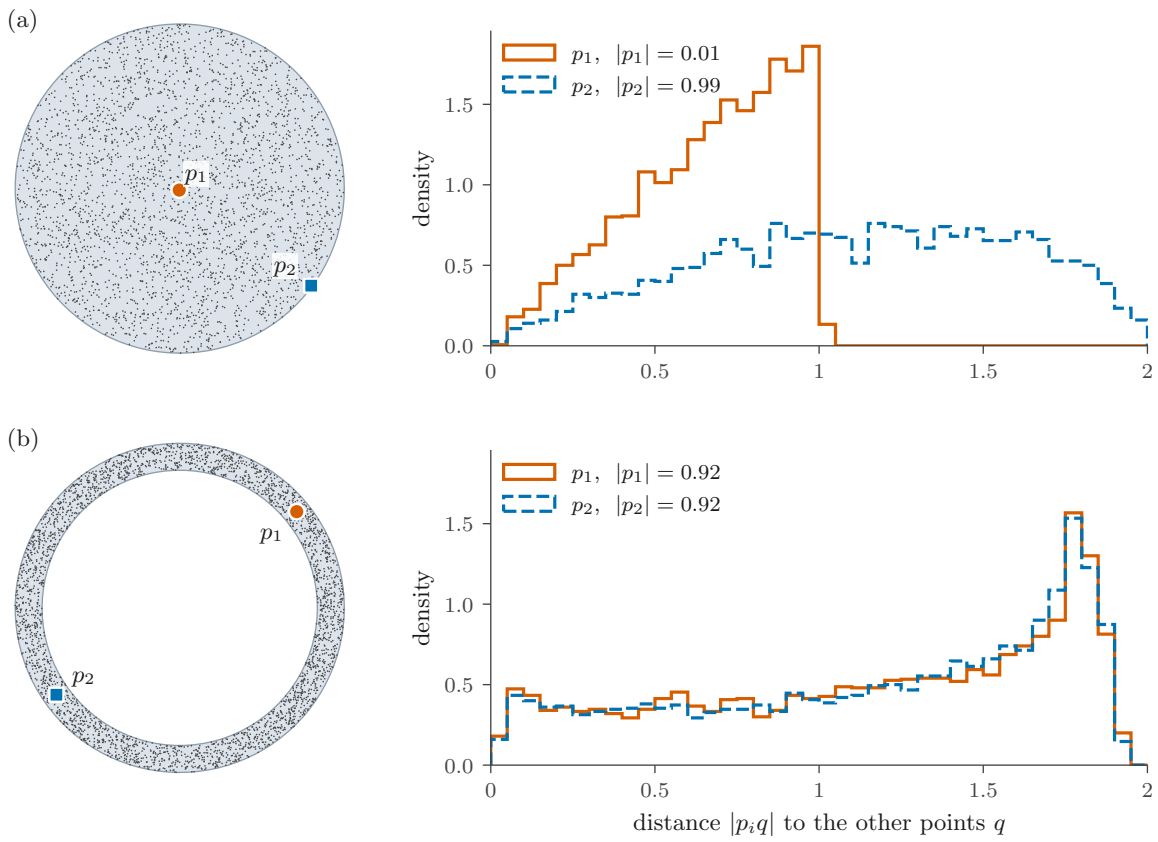


Figure 6: Why the points lie in a ring. (a) In a set spread over a disk, a point near the centre and a point near the edge see very different lists of distances (right: histograms of the distances from each of the two points to all others). (b) In a set spread over a ring, two points at the same distance 0.92 from the centre see nearly the same list. The list of distances from p depends on $|p|$: by (3), the sum of the squared distances is $n|p|^2 + S$. If D is close to $n/3$, every point uses most distances of the set exactly three times, so all points see nearly the same list, and by (3) and (4) they are nearly at the same distance from the centroid. (Random sets with 3000 points; they are illustrations, not sets with few distances.)

For $p \in P$ put

$$m_p = \frac{1}{n} \sum_{r \in R} \mathrm{def}_r(p)\, r^2 \;\geq\; 0,$$

the missing moment of p, divided by n.

Lemma 5. *(N2)* $|p|^2 = 1 - m_p$ *for every* $p \in P$. *(N3)* $|p| \leq 1$, *every distance in* P *is at most* 2, *and* $R \subset (0, 2]$. *(N4) If* $D \leq (1/3 + \delta)n$, *then* $m_p \leq 12\delta + 4/n$ *for every* p.

Proof. (N2): by (3), (4) and (N1), $n|p|^2 + S = 3\sum_r r^2 - nm_p = n + S - nm_p$. (N3) follows from $m_p \geq 0$: all points are in the unit disk, so any two are at distance at most 2. (N4): by (N3) every $r \in R$ satisfies $r^2 \leq 4$, so $m_p \leq 4Q/n$ by (1), and $Q = 3D - n + 1 \leq 3\delta n + 1$. □

So after the normalization, the outer boundary of the ring is the unit circle, and the depth m_p of a point below the unit circle (measured in $|p|^2$) is controlled by its empty slots. A point with all slots full would be on the unit circle. The general position hypothesis enters here only through $d_r(p) \leq 3$.

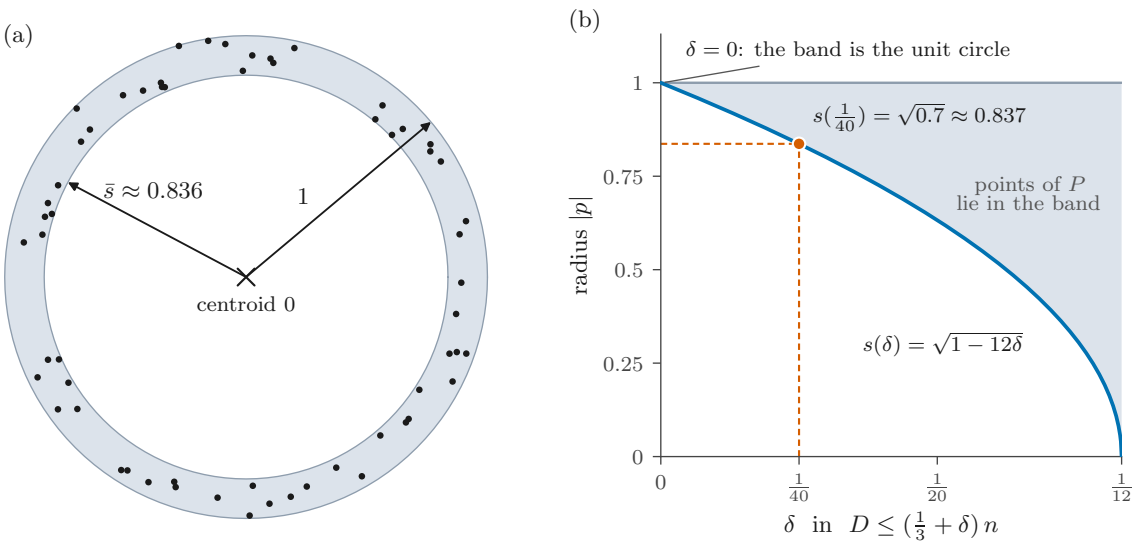


Figure 7: (a) The annulus $\bar{s} \le |p| \le 1$ with $\bar{s} = 41821/50000 \approx 0.836$, drawn to scale. After the normalization of Section 3, every point of a set in general position with $n \ge 10^4$ and $D \le (1/3{+}1/40)n$ lies in it; the centroid of the set is at the centre. The dots are an illustration. (b) The inner radius $\sqrt{1 - 12\delta}$ of the ring for a set with $D \le (1/3 + \delta)n$ (ignoring the term $4/n$). At $\delta = 1/40$ it is $\sqrt{0.7} \approx 0.837$; the proof uses the slightly smaller rational number $\bar{s} \approx 0.836$, which also absorbs the term $4/n$. As $\delta \to 0$ the ring shrinks to the unit circle, and general position does not allow four points on a circle.

The instance. From now on assume, for a contradiction, that $n \ge n_0 = 10^4$ and $D \le (1/3 + \delta)n$ with $\delta = 1/40$. Put $\bar{m} = 12\delta + 4/n_0 = 3/10 + 4/10^4$ and

$$\bar{s} = \frac{41821}{50000} = 0.83642, \qquad \bar{s}^2 = \frac{1748996041}{2500000000} \le 1 - \bar{m}, \qquad \omega := 1 - \bar{s}^2 \approx 0.3004.$$

By Lemma 5, every point of P satisfies

$$\bar{s}^2 \le |p|^2 \le 1. \tag{5}$$

This is *the annulus*: the ring between the circles of radius $\bar{s} \approx 0.836$ and 1 around the centroid (Figure 7(a)). The rational number $\bar{s}$ is chosen slightly below $\sqrt{1 - \bar{m}}$ so that all later computations are with exact fractions. For a smaller δ the ring is thinner (Figure 7(b)).

For $p \in P$ put $y_p = |p|^2 - \bar{s}^2 \in [0, \omega]$, the height of p above the inner boundary (measured in $|p|^2$), and let $\bar{y} = n^{-1} \sum_p y_p$ be its average. The final step of the proof splits into cases according to the value of $\bar{y}$.

4 The distance profile and short chords

The unknowns. For $\rho \ge 0$ let $N_p(\rho) = \#\{q \in P : |pq| \le \rho\}$ (with p included) and $R(\rho) = \#\{r \in R : r \le \rho\}$. Put

$$\mathfrak{d} = \frac{D}{n}, \quad \varepsilon = \frac{1}{n}, \quad \mathfrak{q} = \frac{Q}{n} = 3\mathfrak{d} - 1 + \varepsilon, \quad \mathcal{R}(\rho) = \frac{R(\rho)}{n}, \quad \mathcal{A}(\rho) = \frac{1}{n^2} \sum_p \big(N_p(\rho) - 1\big),$$

and $\mathcal{X}(\rho) = n^{-2} \sum_p \sum_{r \in R,\, r \le \rho} \mathrm{def}_r(p)$. In words:

- $\mathcal{A}(\rho)$ is the fraction of ordered pairs at distance at most ρ: $n^2\mathcal{A}(\rho)$ is the number of ordered pairs of distinct points at distance at most ρ, and the number of unordered pairs e with $|e| \le \rho$ is $n^2\mathcal{A}(\rho)/2$. It is a cumulative distribution function of the pair distances, up to the factor $1-\varepsilon$.
- $\mathcal{R}(\rho)$ is the number of distinct distances up to ρ, divided by n. So $3\mathcal{R}(\rho)$ is the number of slots per point on circles of radius at most ρ, divided by n.
- $\mathcal{X}(\rho)$ is the number of empty slots on circles of radius at most ρ, summed over all points and divided by n^2. Since $\sum_{r\le\rho} d_r(p) = N_p(\rho) - 1$ (filled slots), we have $\mathcal{X} = 3\mathcal{R} - \mathcal{A}$.
- $\mathfrak{d}$, ε and $\mathfrak{q}$ are the number of distances, 1 and the deficit, all divided by n.

The proof treats the values of these functions at finitely many radii as the unknowns of a linear system. Figure 11 in Section 10 shows what a solution of that system looks like when δ is slightly larger than $1/40$.

Facts. (F1) $\mathcal{R}$, $\mathcal{A}$ and $\mathcal{X}$ are nondecreasing, $0 \le \mathcal{X} \le \mathfrak{q}$ and $\mathcal{R} \le \mathfrak{d}$. (F2) $\mathcal{R}(2) = \mathfrak{d}$ and $\mathcal{A}(2) = 1-\varepsilon$, by (N3). (F3) $1/3 - 1/(3n_0) \le \mathfrak{d} \le 1/3 + \delta$ and $0 < \varepsilon \le 1/n_0$.

Short chords. The first geometric consequence of the annulus concerns a circle $C(p,r)$ around a point p of the ring. If the radius r is neither very small nor very large, the circle leaves the ring and comes back, and the part of it inside the ring consists of two arcs, mirror images of each other in the line through 0 and p (Figure 8). (A very small circle can lie entirely inside the ring; then the lemma below gives no useful bound, and the certificate uses the lemma only for radii where it does.) If the circle holds three points of P, two of them are on the same arc, so they are close to each other. The next lemma computes how close. In it, φ is the angle at p, measured from the direction that points away from 0; L and $L+w$ are the values of $\cos\varphi$ at which $C(p,r)$ crosses the inner and the outer boundary circle of the annulus; cl clamps a number to $[-1,1]$; and f is the angle that one arc spans, seen from p.

For $\tau > 0$ and an interval I let $C(I,\tau)$ be the statement: for every $p \in P$ and every $r \in R \cap I$ with $d_r(p) = 3$, two of the three points of P on $C(p,r)$ are at distance at most τ.

Lemma 6 (same-side chord)**.** *Let $p \in P$, $a = |p|$, $r > 0$, $L = (\bar{s}^2 - a^2 - r^2)/(2ar)$, $w = (1-\bar{s}^2)/(2ar)$, let* cl *be the clamp to $[-1,1]$ and $f = \arccos(\mathrm{cl}\, L) - \arccos(\mathrm{cl}(L+w)) \in [0,\pi]$. Then any three points of P on $C(p,r)$ contain two at squared distance at most $2r^2(1-\cos f)$.*

Proof. Rotate so that $p = (a,0)$ and write $q = p + r(\cos\varphi, \sin\varphi)$. Then $|q|^2 = a^2 + r^2 + 2ar\cos\varphi$, so by (5) a point $q \in P \cap C(p,r)$ has $\cos\varphi \in [L, L+w]$. Two of three points lie in the same closed half $\varphi \in [0,\pi]$ or $\varphi \in [\pi, 2\pi]$; reflecting both in the first axis if needed (this preserves their distance), both lie in $[0,\pi]$. Since cos is a decreasing bijection $[0,\pi] \to [-1,1]$, both angles lie in $[\arccos \mathrm{cl}(L+w), \arccos \mathrm{cl}\, L]$, so they differ by at most f, and the two points are at distance $2r\sin(|\Delta\varphi|/2) \le 2r\sin(f/2)$. □

The bound of Lemma 6 depends on $a = |p|$ and r. A computer can only check finitely many cases, so the next lemma gives one bound for a whole box of values $a \in [a_0, a_1]$, $r \in [r_0, r_1]$. Covering the annulus and a range of radii by finitely many boxes then proves a statement $C(I,\tau)$.

Lemma 7 (box bound)**.** *Let $\bar{s} \le a_0 < a_1 \le 1$ and $0 < r_0 < r_1$. Put $L_{\mathrm{lo}} = (\bar{s}^2 - a_1^2 - r_1^2)/(2a_0r_0)$, $L_{\mathrm{hi}} = (\bar{s}^2 - a_0^2 - r_0^2)/(2a_1r_1)$, $w_{\mathrm{hi}} = (1-\bar{s}^2)/(2a_0r_0)$,*

$$\Lambda = \{L_{\mathrm{lo}}, L_{\mathrm{hi}}\} \cup \big(\{-1,\, 1 - w_{\mathrm{hi}}\} \cap [L_{\mathrm{lo}}, L_{\mathrm{hi}}]\big),$$

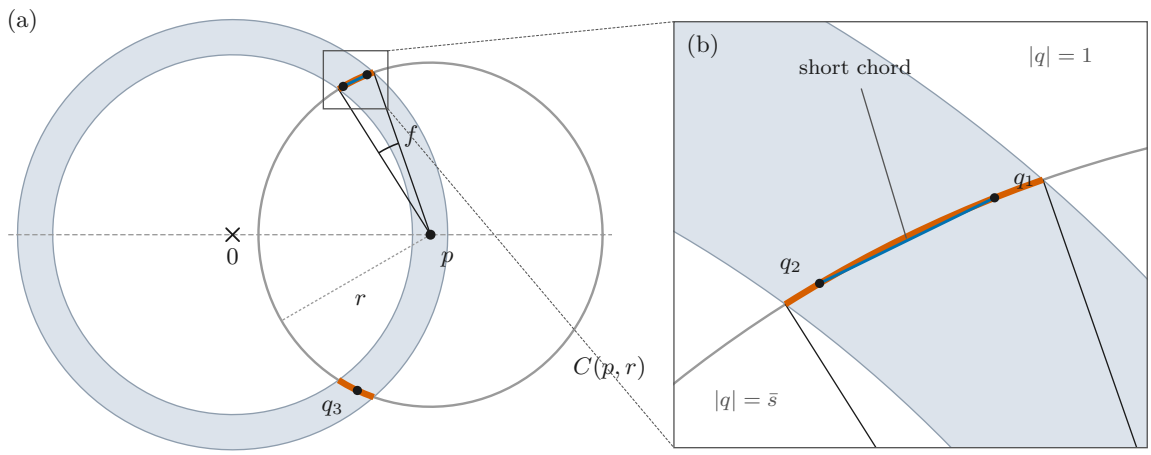


Figure 8: Lemma 6. (a) A circle $C(p, r)$ of medium radius around a point p of the annulus meets the annulus in two arcs (thick), mirror images in the line through the centroid 0 and p. Seen from p, each arc spans the angle f. Of three points q_1, q_2, q_3 of P on the circle, two lie on the same arc. (b) A zoom on the upper arc: q_1 and q_2 are joined by a short chord. Here $|p| = 0.92$, $r = 0.8$ and $f \approx 13°$.

and let $\beta \le \min_{L\in\Lambda}\left[\operatorname{cl} L\cdot\operatorname{cl}(L+w_{\mathrm{hi}})+\sqrt{(1-(\operatorname{cl} L)^2)(1-(\operatorname{cl}(L+w_{\mathrm{hi}}))^2)}\right]$. *Then for every* $a \in [a_0, a_1]$ *and* $r \in [r_0, r_1]$ *the bound of Lemma 6 is at most* $2r_1^2(1-\beta)$. *Also, for every* $r \in (0, r_1]$, *any two points of* $C(p, r)$ *are at squared distance at most* $4r_1^2$.

Proof. The numerator $\bar{s}^2 - a^2 - r^2$ is negative and the denominator $2ar$ is positive, so $L \in [L_{\mathrm{lo}}, L_{\mathrm{hi}}]$ and $w \le w_{\mathrm{hi}}$. For fixed L, $f(L, w)$ is nondecreasing in w. For fixed w, $f(\cdot, w)$ is continuous and is: 0 if $L + w \le -1$; increasing when $L < -1 < L + w \le 1$; on $-1 \le L$, $L + w \le 1$ its derivative $-1/\sqrt{1-L^2} + 1/\sqrt{1-(L+w)^2}$ is ≤ 0 for $L \le -w/2$ and ≥ 0 for $L \ge -w/2$, so it decreases and then increases; decreasing when $-1 \le L \le 1 < L + w$; 0 if $L > 1$; and π on $[1-w, -1]$ when $w > 2$. When $w \le 2$ we have $-1 \le -w/2 \le 1 - w$, so the only interior local maxima are at -1 and $1 - w$. Hence the maximum of $f(\cdot, w_{\mathrm{hi}})$ on $[L_{\mathrm{lo}}, L_{\mathrm{hi}}]$ is attained at a point of Λ. Finally $\cos(\arccos u - \arccos v) = uv + \sqrt{1-u^2}\sqrt{1-v^2}$, so $\cos f \ge \beta$ on the box, and $2r^2(1-\cos f) \le 2r_1^2(1-\beta)$. The last claim holds because $C(p, r)$ has diameter $2r$. □

In the certificate, every statement $C(I, \tau)$ that an inequality needs is proved by covering $[\bar{s}, 1] \times \overline{I}$ with finitely many boxes, on each of which Lemma 7 gives a bound at most τ^2, with square roots rounded so that β is a lower bound; a box with $r_0 = 0$ uses the bound $4r_1^2$. When no such cover exists for an interval, the corresponding inequality is not used.

5 Pair counts, apex ranges, a cap and a moment

This section derives four families of inequalities for the profile.

What the next lemma says. Count, for each point p, the pairs $\{q, q'\}$ of points on a common circle around p that are at distance at most ρ from each other. A pair is counted once for each of its apexes, so at most twice, and the total count is at most twice the number of close pairs, which is $n^2\mathcal{A}(\rho)$. On the other hand, a circle of radius at most $\rho/2$ has diameter at most ρ, so all of its pairs are close; and a full circle of radius in $(\rho/2, r']$ has a close pair by the chord statement. Comparing

the two counts gives lower bounds for $\mathcal{A}(\rho)$, the fraction of close pairs, in terms of the profile at other radii.

Lemma 8 (pair counts)**.** *Let $\rho > 0$ and $\rho/2 < r' \le 2$, and assume $C((\rho/2, r'], \rho)$. Then*

$$\begin{aligned} &(a) \quad \mathcal{A}(\rho) \ge \mathcal{A}(r') - 2\mathcal{R}(r'), \\ &(b) \quad \mathcal{A}(\rho) \ge \mathcal{A}(\rho/2) + \mathcal{A}(r') - \mathcal{R}(\rho/2) - 2\mathcal{R}(r'), \\ &(c) \quad \mathcal{A}(\rho) \ge \mathcal{A}(r') - \mathcal{A}(\rho/2) - 2\mathcal{R}(r') + 2\mathcal{R}(\rho/2). \end{aligned}$$

Proof. Let $I_p = \sum_{r\in R} \#\{\{q,q'\} \subset P \cap C(p,r) : |qq'| \le \rho\}$. An unordered pair $\{q,q'\}$ with $|qq'| \le \rho$ is counted once for each apex p, that is $\alpha(q,q') \le 2$ times, so $\sum_p I_p \le 2\#\{\{q,q'\} : |qq'| \le \rho\} = n^2\mathcal{A}(\rho)$. A radius $r \le \rho/2$ contributes $\binom{d_r(p)}{2}$ to I_p, because all pairs on $C(p,r)$ are within $2r \le \rho$. A radius $r \in (\rho/2, r']$ contributes at least $[d_r(p) = 3] \ge d_r(p) - 2$ by the hypothesis, where $[d_r(p) = 3]$ is 1 if $d_r(p) = 3$ and 0 otherwise. For $d \in \{0,1,2,3\}$, $\binom{d}{2} \ge d - 2$, $\binom{d}{2} \ge 2d - 3$ and $\binom{d}{2} \ge 0$. Dropping the radii above r' and using $\sum_{r\le y} d_r(p) = N_p(y) - 1$, the three choices, used for the radii $r \le \rho/2$, give in this order, for each p, $I_p \ge N_p(r') - 1 - 2R(r')$, $I_p \ge (N_p(\rho/2) - 1) + (N_p(r') - 1) - R(\rho/2) - 2R(r')$, and $I_p \ge N_p(r') - N_p(\rho/2) - 2R(r') + 2R(\rho/2)$. Sum over p and divide by n^2. □

What the next lemma says. Take a long pair $\{a,b\}$, with $|ab| > \lambda$, and an apex z of it. All three points lie in the annulus, and this restricts the distance $r = |za|$: its square lies between two explicit numbers $\mathrm{lo}(\lambda)$ and $\mathrm{hi}(\lambda)$. By Lemma 2, most long pairs have two apexes. So the set must have many pairs (z,a) with distance in the allowed range, and this is a lower bound for the increase of $\mathcal{A}$ over that range.

Lemma 9 (apex range)**.** *Let $0 < \lambda < 2$ with $\lambda^2 \ge 4\omega$, put $\mathrm{lo}(\lambda) = \bar{s}^2 + 1 - \bar{s}\sqrt{4-\lambda^2}$ and $\mathrm{hi}(\lambda) = 2 + \sqrt{4-\lambda^2}$, take $0 < w^2 \le \mathrm{lo}(\lambda)$ and $\mathrm{hi}(\lambda) \le h^2 \le 4$, and assume $C((w,2],\lambda)$. Then*

$$2\big(1 - \varepsilon - \mathcal{A}(\lambda)\big) - \tfrac{4}{3}\big(\mathcal{A}(h) - \mathcal{A}(w)\big) \;\le\; 2\mathfrak{q}.$$

Proof. (i) Let $z \notin \{a,b\}$ with $|za| = |zb| = r$ and $|ab| > \lambda$. Put $u = |a|^2 + |b|^2$ and $\zeta = |z|$. The parallelogram law gives $|a+b|^2 = 2u - |ab|^2 < 2u - \lambda^2$, and $2r^2 = |z-a|^2 + |z-b|^2 = 2\zeta^2 + u - 2\langle z, a+b\rangle$. Hence $\Psi^- < 2r^2 < \Psi^+$ with $\Psi^\pm = 2\zeta^2 + u \pm 2\zeta\sqrt{2u-\lambda^2}$. By (5), $\zeta \in [\bar{s}, 1]$ and $u \in [2\bar{s}^2, 2]$. Then $\partial_\zeta \Psi^- \ge 4\bar{s} - 2\sqrt{4-\lambda^2} \ge 0$ and, at $\zeta = \bar{s}$, $\partial_u \Psi^- = 1 - 2\bar{s}/\sqrt{2u-\lambda^2} \le 0$ for $u \in (\lambda^2/2, 2]$, both because $4\bar{s}^2 \ge 4 - \lambda^2$; the actual u lies in $(\lambda^2/2, 2]$ because $0 \le |a+b|^2 < 2u - \lambda^2$, and Ψ^- is continuous on $[\lambda^2/2, 2]$. So $\Psi^- \ge \Psi^-(\bar{s}, 2) = 2\,\mathrm{lo}(\lambda)$, and $\Psi^+ \le 4 + 2\sqrt{4-\lambda^2} = 2\,\mathrm{hi}(\lambda)$.

(ii) Let Φ be the set of ordered pairs (a,b) with $|ab| > \lambda$; $|\Phi| = n^2(1 - \varepsilon - \mathcal{A}(\lambda))$. Each apex z of $(a,b) \in \Phi$ lies at a distance r from a with $\mathrm{lo} < r^2 < \mathrm{hi}$ by (i), so $w < r \le h$. A circle $C(z,r)$ with d points of P carries at most $\frac{4}{3}d$ ordered pairs of Φ: none for $d \le 1$, two for $d = 2$, and for $d = 3$ at most four, because $C((w,2],\lambda)$ gives a pair within λ. Hence $\sum_\Phi \alpha \le \frac{4}{3}\sum_z \#\{q : w < |zq| \le h\} = \frac{4}{3}n^2(\mathcal{A}(h) - \mathcal{A}(w))$. Insert this into (2), $2|\Phi| - \sum_\Phi \alpha \le 2nQ$, and divide by n^2. □

What the next lemma says. The centroid is at the origin, so the points cannot all crowd on one side of it. Around each point z, a weight that is large near z and averages to 1 over the set then shows that not too many points are close to z. The result is an upper bound for a weighted average of $\mathcal{A}$ over the radii up to $\rho_e \approx 1.75$.

Lemma 10 (averaged cap)**.** *Let $\rho_e^2 = 2\bar{s}^2 + 2\bar{s}$. For $0 < g_1 < \cdots < g_m \le \rho_e$, $\sum_{i<m} \mathcal{A}(g_i)(g_{i+1}^2 - g_i^2) \le 2\bar{s}$.*

Proof. Fix $z \in P$, $\zeta = |z| \in [\bar{s}, 1]$, and put $F(q) = 1 + \langle q, z/\zeta \rangle$. By (N3), $F \geq 0$ on P, and $\sum_q F(q) = n$ because the centroid is 0. Since $\langle q, z/\zeta \rangle = (|q|^2 + \zeta^2 - |q - z|^2)/(2\zeta)$, (5) gives $F(q) \geq \varphi_\zeta(|q - z|)$ with $\varphi_\zeta(t) = 1 + (\bar{s}^2 + \zeta^2 - t^2)/(2\zeta)$. Its derivative in ζ is $\frac{1}{2} - (\bar{s}^2 - t^2)/(2\zeta^2) \geq 0$ for $\zeta \geq \bar{s}$, so $\varphi_\zeta \geq \varphi_{\bar{s}} = (\rho_e^2 - t^2)/(2\bar{s})$. Therefore $n \geq \sum_q \max(0, \varphi_{\bar{s}}(|qz|)) = \frac{1}{\bar{s}} \int_0^{\rho_e} t\, N_z(t)\, dt$. Summing over z, $\int_0^{\rho_e} t\,(n^2 \mathcal{A}(t) + n)\, dt \leq \bar{s} n^2$, so $\int_0^{\rho_e} t \mathcal{A}(t)\, dt \leq \bar{s}$. Since $\mathcal{A}$ is nondecreasing, the left side of the claim is at most twice this integral. □

What the next lemma says. The average of $|pq|^2$ over all ordered pairs (p, q) is $2S/n$ by (3), and by (N2) S/n, the average of $|p|^2$, is 1 minus the average missing moment. Written in terms of the profile, this gives an exact identity between $\mathcal{A}$, $\mathcal{X}$ and the deficit $\mathfrak{q}$. Since the profile is only known at grid points, the proof uses it as two inequalities (lower and upper Riemann sums). The proof writes the average missing moment as the Stieltjes integral $\int_0^2 \rho^2\, d\mathcal{X}(\rho)$, which here is the finite sum, over $r \in R$, of r^2 times the jump of $\mathcal{X}$ at r, and then integrates by parts.

Lemma 11 (second moment)**.** $\int_0^2 \rho \big(\mathcal{A}(\rho) + 2\mathcal{X}(\rho)\big)\, d\rho = 1 + 4\mathfrak{q} - 2\varepsilon$. *The integrand is nondecreasing by (F1), so on a partition* $0 = g_0 < \cdots < g_m = 2$ *the sum* $\sum_i \frac{1}{2}(g_{i+1}^2 - g_i^2)(\mathcal{A} + 2\mathcal{X})(g_i)$ *is at most* $1 + 4\mathfrak{q} - 2\varepsilon$*, and the same sum with* $(\mathcal{A} + 2\mathcal{X})(g_{i+1})$ *is at least* $1 + 4\mathfrak{q} - 2\varepsilon$.

Proof. For each ordered pair (p, q), $p = q$ allowed, $2 - |pq|^2/2 = \int_{|pq|}^2 \rho\, d\rho$ by (N3). Summing and dividing by n^2: the left side becomes $\int_0^2 \rho(\mathcal{A} + \varepsilon)\, d\rho$, and the right side becomes $2 - S/n = 1 + m_{\text{avg}}$ with $m_{\text{avg}} = n^{-1} \sum_p m_p$, by $\sum_{p,q} |p - q|^2 = 2nS$ and (N2). Finally $m_{\text{avg}} = \int_0^2 \rho^2\, d\mathcal{X}(\rho) = 4\mathcal{X}(2) - 2\int_0^2 \rho \mathcal{X}(\rho)\, d\rho$ with $\mathcal{X}(2) = \mathfrak{q}$ by (N3) and (1). □

6 Apex bands

What this section does. Lemma 9 uses the worst case over all positions in the annulus. This section keeps more information. For an isosceles triangle (z, a, b) with a long base ab, the geometry of the annulus confines the apex distance $r = |za|$ to a range whose ends move linearly with the heights $y_a + y_b$ and y_z of the three points above the inner boundary. Splitting the triangles into bands by the value of r and recording the total heights in each band gives linear inequalities whose coefficients involve $\bar{y}$, the average height. Part (AB5) is the only place where $\bar{y}$ multiplies an unknown; this is why the final step splits into cases by $\bar{y}$.

Fix a rational $\lambda < 2$ such that $C((0, 2], \lambda)$ holds. Let Φ be the set of ordered pairs with $|ab| > \lambda$ and $T = \{(z, a, b) : (a, b) \in \Phi,\ z \notin \{a, b\},\ |za| = |zb|\}$, the isosceles triangles with a long base. Fix $e_0 < e_1 < \cdots < e_m = 2$ with $e_0 \leq \lambda/2$, let T_j be the triangles with $e_j < |za| \leq e_{j+1}$, and put $W_j = |T_j|/n^2$, $U_j = n^{-2} \sum_{T_j} (y_a + y_b)$, $V_j = n^{-2} \sum_{T_j} y_z$, and for $t > 0$, $K^{\pm}(t) = (4 \pm t)\bar{s}^2 \pm (4\bar{s}^2 - \lambda^2)/t$. Here $t > 0$ is a free parameter, $x^+ = \max(x, 0)$, and W_j, U_j, V_j depend on λ and on the band edges.

Lemma 12 (apex bands)*(AB0)* $\bar{y} = \omega - 4\mathfrak{q} + 2\int_0^2 \rho\, \mathcal{X}(\rho)\, d\rho$.

(AB1) $\sum_j W_j \geq 2(1 - \varepsilon - \mathcal{A}(\lambda)) - 2\mathfrak{q}$.

(AB2) $W_j \leq \frac{4}{3}\big(\mathcal{A}(e_{j+1}) - \mathcal{A}(e_j)\big)$.

(AB3) $(1 + 2/t)U_j + (2 + t)V_j \geq (2e_j^2 - K^+(t))W_j$.

(AB4) $(2/t - 1)^+ U_j + (t - 2)^+ V_j \geq (K^-(t) - 2e_{j+1}^2)W_j$.

(AB5) $\sum_j U_j \leq 12\, \bar{y}\, (\mathfrak{d} - \mathcal{R}(\lambda))$ *and, for every* $h \leq \lambda/2$, $\sum_j V_j \leq 4\, \bar{y}\, (\mathfrak{d} - \mathcal{R}(h))$.

Proof. Every triangle of T has $2|za| \geq |ab| > \lambda$ and $|za| \leq 2$, so it lies in exactly one band. (AB0): $n^{-1}\sum_p |p|^2 = 1 - m_{\mathrm{avg}}$ by (N2), and $m_{\mathrm{avg}} = 4\mathfrak{q} - 2\int_0^2 \rho\mathcal{X}$ as in the proof of Lemma 11. (AB1): $|T| = \sum_\Phi \alpha \geq 2|\Phi| - 2nQ$ by (2). (AB2): fix z and a radius r in the band; $C(z,r)$ carries at most $\frac{4}{3}d_r(z)$ ordered pairs of Φ, as in Lemma 9(ii), now using $C((0,2],\lambda)$; sum over the radii in the band and over z.

For (AB3) and (AB4) take $(z,a,b) \in T$, $r = |za|$, $\ell = |ab|$, and write $|z|^2 = \bar{s}^2 + y_z$, $u = |a|^2 + |b|^2 = 2\bar{s}^2 + Y$ with $Y = y_a + y_b$, and $\Theta = 2u - \lambda^2 = 4\bar{s}^2 - \lambda^2 + 2Y$. As in Lemma 9, $2r^2 = 2|z|^2 + u - 2\langle z, a+b\rangle$ and $|a+b|^2 = 2u - \ell^2 < \Theta$, so $\Theta > 0$ and, by the Cauchy–Schwarz and AM–GM inequalities, $|2\langle z, a+b\rangle| < 2\sqrt{|z|^2\Theta} \leq t|z|^2 + \Theta/t$. Hence

$$K^-(t) + (1-2/t)Y + (2-t)y_z \;<\; 2r^2 \;<\; K^+(t) + (1+2/t)Y + (2+t)y_z.$$

On T_j, $r > e_j$; summing the right inequality over T_j gives (AB3). On T_j, $r \leq e_{j+1}$, so $(2/t-1)Y + (t-2)y_z > K^-(t) - 2e_{j+1}^2$; replace the coefficients by their positive parts ($Y, y_z \geq 0$) and sum, which gives (AB4).

(AB5): Φ is symmetric, so $\sum_T (y_a + y_b) = 2\sum_{(a,b)\in\Phi} \alpha(a,b)\, y_a \leq 4\sum_a y_a M_a$ with $M_a = \#\{b : |ab| > \lambda\} = \sum_{r>\lambda} d_r(a) \leq 3(D - R(\lambda))$. The ordered pairs of Φ with apex z lie on circles $C(z,r)$ with $r > \lambda/2 \geq h$, and by the count in (AB2) there are at most $\frac{4}{3}\sum_{r>h} d_r(z) \leq 4(D - R(h))$ of them. Multiply by y_z, sum, and divide by n^2. □

7 The bisector length gap

The inequalities so far use only the number α of apexes of each pair. The next lemma uses where the apexes are.

What the next lemma says. Take a long pair $f = \{a,b\}$. Both endpoints lie in the thin annulus, so they are at nearly the same distance from the centroid, and the perpendicular bisector of ab passes close to the centroid. A line through (or near) the centre of a ring crosses the ring in two short segments on opposite sides of the centre (Figure 9). The two apexes of f are points of P, so they lie in the ring, on these two segments. Either both are on the same segment, and then they are close to each other, or they are on opposite segments, and then they are far apart. Medium distances are impossible. Since κ is injective (Lemma 3), this gives: the number of long pairs with two apexes is at most the number of short pairs plus the number of very long pairs.

Lemma 13 (bisector length gap)**.** *Let $X \subset \mathbb{R}^2$ be finite, no three points collinear and no four concyclic, with $\bar{s}^2 \leq |p|^2 \leq 1$ for all $p \in X$, where $0 < \bar{s} < 1$, and let $\omega = 1 - \bar{s}^2$; define $\mathcal{D}$, κ, N and x_2 for X as in Section 2. Fix $\lambda > 0$ with $h_0 := \omega/(2\lambda) < \bar{s}$ and put*

$$\sigma_0 = \sqrt{1-h_0^2} - \sqrt{\bar{s}^2 - h_0^2}, \qquad \Lambda_0 = 2\sqrt{\bar{s}^2 - h_0^2}.$$

(i) If $f \in \mathcal{D}$ and $|f| \geq \lambda$, then $|\kappa(f)| \leq \sigma_0$ or $|\kappa(f)| \geq \Lambda_0$.

(ii) $\#\{e : \lambda \leq |e| < \Lambda_0\} \leq \#\{e : |e| \leq \sigma_0\} + (N - x_2)$, where e runs over unordered pairs.

(No centroid condition is used; the lemma is applied to $X = P$.)

Proof. (i) For $f = \{a,b\}$, $\mathrm{bis}(f) = \{x : 2\langle x, b-a\rangle = |b|^2 - |a|^2\}$ lies at distance $h = \big||b|^2 - |a|^2\big|/(2|ab|) \leq \omega/(2\lambda) = h_0 < \bar{s}$ from 0. Write $\mathrm{bis}(f) = \{c + tu : t \in \mathbb{R}\}$ with c the foot of the perpendicular from 0 and $|u| = 1$, so $|c+tu|^2 = h^2 + t^2$. The two points $z = c + tu$, $z' = c + t'u$

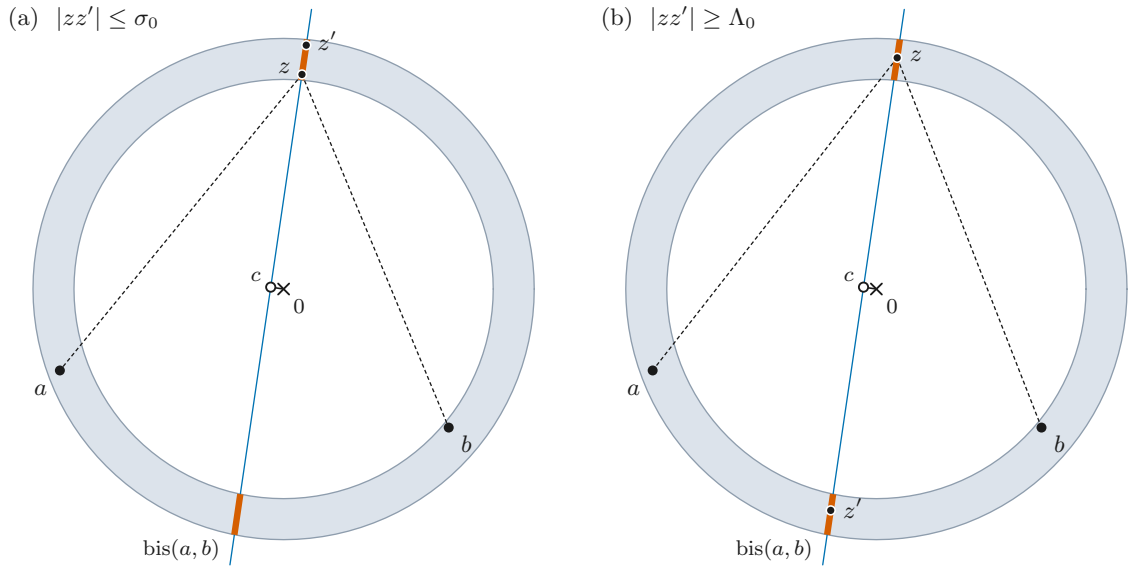


Figure 9: Lemma 13 with $\lambda = 1.2$. The perpendicular bisector of a long pair ab passes within $h_0 = \omega/(2\lambda)$ of the centroid, and meets the annulus in two segments (thick). The two apexes z, z' of ab are points of P, so they lie on these segments. (a) On the same segment, $|zz'| \le \sigma_0$. (b) On opposite segments, $|zz'| \ge \Lambda_0$.

of $\kappa(f)$ satisfy $\bar{s}^2 \le h^2 + t^2, h^2 + t'^2 \le 1$, so $\sqrt{\bar{s}^2 - h^2} \le |t|, |t'| \le \sqrt{1 - h^2}$; in particular t and t' are nonzero. If $tt' > 0$, then $|zz'| = \big||t| - |t'|\big| \le \sigma(h) := \sqrt{1 - h^2} - \sqrt{\bar{s}^2 - h^2}$. If $tt' < 0$, then $|zz'| = |t| + |t'| \ge \Lambda(h) := 2\sqrt{\bar{s}^2 - h^2}$. On $[0, \bar{s})$, $\sigma'(h) = h\big(1/\sqrt{\bar{s}^2 - h^2} - 1/\sqrt{1 - h^2}\big) \ge 0$ and Λ is decreasing, so $\sigma(h) \le \sigma_0$ and $\Lambda(h) \ge \Lambda_0$.

(ii) Let $F = \{f \in \mathcal{D} : |f| \ge \lambda\}$. By (i) and Lemma 3, $|F| \le \#\{|e| \le \sigma_0\} + \#\{|e| \ge \Lambda_0\}$. Every pair with $|e| \ge \lambda$ that is not in F is outside $\mathcal{D}$, so $|F| \ge \#\{|e| \ge \lambda\} - (N - x_2)$. Subtracting gives (ii). (If $\Lambda_0 \le \lambda$, the left side of (ii) is 0.) □

In the language of the profile, Lemma 13 becomes a linear inequality:

Corollary 14 (row J)**.** *In the setting of Section 3, let λ be as in Lemma 13, and let $g_2 < \Lambda_0$ and $g_\sigma \ge \sigma_0$ with $g_2 > \lambda$. Then $\mathcal{A}(g_2) - \mathcal{A}(\lambda) - \mathcal{A}(g_\sigma) \le 2\mathfrak{q}$.*

Proof. $\{\lambda < |e| \le g_2\} \subseteq \{\lambda \le |e| < \Lambda_0\}$ and $\{|e| \le \sigma_0\} \subseteq \{|e| \le g_\sigma\}$. Apply Lemma 13(ii) and $N - x_2 \le nQ$ (Lemma 2), multiply by $2/n^2$, and use $\#\{|e| \le \rho\} = n^2\mathcal{A}(\rho)/2$. □

Remark 15. Corollary 14 adds information to the other families. A closely related system without it (the same families with 34 thresholds and a slightly different set of (AB5)-type rows, built at $\delta = 1/44$ and at $\delta = 177/7832$) has exact rational feasible points, and these points violate Corollary 14 by about 0.2827 and 0.3045 (exact rational evaluation). At $\delta = 1/40$ the system of Section 8 without Corollary 14 is feasible in floating-point arithmetic.

8 The certificate

This section turns the lemmas into a finite system of linear inequalities and explains how its infeasibility is certified.

Certificates of infeasibility. Suppose a system of linear inequalities $Av \le b$ (one inequality per row of the matrix A) has a solution v, and let $y \ge 0$ be a vector of nonnegative multipliers, one per row. Multiplying row i by y_i and adding gives $(y^{\mathsf{T}}A)\,v \le y^{\mathsf{T}}b$. So if we find $y \ge 0$ for which this combined inequality fails for every possible v, the system has no solution. In the example of Figure 4 the combined inequality is $0 \le -1$. In our system the unknowns are also known to lie in a box (each unknown between known bounds), and it is enough that the combined inequality fails for every v in the box. This is easy to check: the minimum of a linear function over a box is attained coordinatewise. Farkas' lemma says that a certificate exists whenever a system has no solution, but the proof below does not use Farkas' lemma: it only uses the certificate, and the easy argument above.

Why a grid and why cases. The profile functions $\mathcal{A}$ and $\mathcal{R}$ have infinitely many values, but the inequalities only need their values at the finitely many radii of a grid G; the monotonicity facts (F1) connect neighbouring grid points. Most inequalities are linear in these values. The exception is (AB5), where the unknown $\bar{y}$ multiplies the unknowns $\mathfrak{d} - \mathcal{R}(\lambda)$ and $\mathfrak{d} - \mathcal{R}(h)$. Fixing $\bar{y}$ to a short interval and replacing it by the right endpoint of that interval makes (AB5) linear, and (AB0) becomes two linear inequalities that put the right side of (AB0) into the interval. So the proof splits the possible values of $\bar{y}$ into 60 intervals and gives a separate certificate for each.

Grid. Let G be the set of rationals $b/2^j \ge 1/125$ ($j \ge 0$), where b runs over the 160 numbers $2 \cdot 2^{-k/160}$ ($0 \le k < 160$) rounded to six decimals, the number $\lceil \sqrt{3} \cdot 10^6 \rceil / 10^6$, and 2. Then $|G| = 1283$.

Variables and box. The unknowns are $\mathfrak{d}$, ε, $\mathcal{R}(g)$ and $\mathcal{A}(g)$ for $g \in G$, and the band quantities W_j, U_j, V_j of Section 6 for 32 thresholds λ (below). The true values satisfy the box: (F3), $\mathcal{R}(g) \in [0, 1/3 + \delta]$, $\mathcal{A}(g) \in [0,1]$, $W_j \in [0,2]$ and $U_j, V_j \in [0,4]$ (since $|T| = \sum_\Phi \alpha \le 2n^2$, $U_j \le 2\omega W_j$ and $V_j \le \omega W_j$).

Rows. All rows are linear inequalities in these unknowns whose coefficients are explicit rationals; each instance carries its parameters, and every side condition is checked in exact arithmetic. (Each row is stored with a *tag* that names its family and its parameters, for example the radii ρ and r' of Lemma 8. Its *side conditions* are the hypotheses of the lemma for these parameters, such as $C((\rho/2, r'], \rho)$. The *chord table* is the list of the statements $C(I, \tau)$ that the box covers of Lemma 7 prove.)

1. (F1), (F2) on G: monotonicity of $\mathcal{R}$, $\mathcal{A}$ and $\mathcal{X} = 3\mathcal{R} - \mathcal{A}$ between consecutive grid points, $\mathcal{X} \ge 0$, $\mathcal{X} \le \mathfrak{q}$, $\mathcal{R} \le \mathfrak{d}$, $\mathcal{R}(2) = \mathfrak{d}$, $\mathcal{A}(2) = 1 - \varepsilon$.

2. Lemma 8 (a)–(c) for each $\rho \in G$ for which a grid point $r' > \rho/2$ with $C((\rho/2, r'], \rho)$ is certified by Lemma 7 (579 values of ρ; r' is read from the certifier's chord table, and $r' = 2$ when the whole range is certified); (b) and (c) only when $\rho/2 \in G$.

3. Lemma 9 for every $\lambda \in G$ with $4\omega \le \lambda^2 < 4$, w the largest grid point with $w^2 \le \mathrm{lo}(\lambda)$ (rounded down) and h the smallest grid point with $h^2 \ge \mathrm{hi}(\lambda)$ (rounded up), when $C((w, 2], \lambda)$ is certified.

4. Lemma 10 on the grid points up to ρ_e (rounded down), and the lower and upper Riemann sums of Lemma 11 on G.

These are 9577 rows. Then:

5. Lemma 12, (AB1)–(AB4), for 32 thresholds $\lambda \in G \cap [1, 2\bar{s} + 1/20)$, each with $C((0,2],\lambda)$ certified, chosen greedily upward from 1 with consecutive thresholds at least $1/50$ apart; band edges $e_0 =$ the largest grid point $\le \lambda/2$ and $e_k =$ the largest grid point $\le \min(e_0 + k/50, 2)$, ending at 2; and $t \in \{(1/50)(5/4)^k : 0 \le k < 25\}$, a row of (AB3) or (AB4) being used only when its coefficient of W_j is positive. These are 15182 rows. (The certificate file also lists 64 rows of a weaker form of (AB5) with weight zero.)

6. Corollary 14 for every $\lambda \in G$ with $h_0 < \bar{s}$: g_2 is the largest grid point below a rational lower bound of Λ_0, g_σ the smallest grid point at or above a rational upper bound of σ_0, and $g_2 > \lambda$. These are 515 rows.

Case split. $\bar{y} \in [0, \omega]$. Split $[0, \omega]$ into the 60 intervals $[Y_k, Y_{k+1}] = \omega\,[k/60, (k+1)/60]$. In case k ($\bar{y} \in [Y_k, Y_{k+1}]$) add, for each threshold λ with $h =$ the largest grid point $\le \lambda/2$,

$$\textstyle\sum_j U_j \le 12\, Y_{k+1}(\mathfrak{d} - \mathcal{R}(\lambda)), \qquad \sum_j V_j \le 4\, Y_{k+1}(\mathfrak{d} - \mathcal{R}(h)),$$

which follow from (AB5) and $\mathfrak{d} - \mathcal{R} \ge 0$, and the two rows

$$Y_k \le \omega - 4\mathfrak{q} + 2\,\mathrm{Up}, \qquad \omega - 4\mathfrak{q} + 2\,\mathrm{Lo} \le Y_{k+1},$$

where Up and Lo are the upper and lower Riemann sums of $\int_0^2 \rho\mathcal{X}$ on G (Up includes the cell $[0, \min G]$ with the value $\mathcal{X}(\min G)$, Lo omits it); they follow from (AB0) because $\mathcal{X}$ is nondecreasing and nonnegative. The variable $\bar{y}$ itself does not occur.

The certificate step. Write the rows of case k as $Av \le b$, where v is the vector of true values. For each k the certificate gives nonnegative rational multipliers y with

$$\min_{v \in \mathrm{box}} (y^{\mathsf{T}} A)\, v - y^{\mathsf{T}} b \;>\; 0.$$

The minimum of a linear function over a box is computed coordinatewise, in exact arithmetic. But the true v lies in the box and satisfies $Av \le b$, so $(y^{\mathsf{T}}A)v \le y^{\mathsf{T}}b$: a contradiction. We call the left side the *gap* of case k. Each case uses 68 to 130 instances of Corollary 14, and the 60 cases use 125126 multiplier rows in all. Figure 10 shows the gap of every case. The smallest gap is in case 30 ($\bar{y} \in [0.1502, 0.1552]$):

$$\frac{427343548856324105093106227147693665008 59}{131072000000000000000000000000000000000000000000000} \;\approx\; 3.2604 \cdot 10^{-6}.$$

Since every case is contradictory and $\bar{y}$ lies in one of them, the assumption $D \le (1/3 + 1/40)n$ with $n \ge 10^4$ is false. This proves Theorem 1. □

9 Verification

Exact verifier. The multipliers were found by a floating-point linear-programming solver and then rounded to rationals; the solver plays no part in the proof. A separate verifier, written with the Python standard library only and not importing the program that found the multipliers, rebuilds every row from its tag, re-checks every side condition (grid membership of each parameter, the chord covers of Lemma 7, the square-root bounds in Lemma 9 and Corollary 14, the cut of Lemma 10, the band edges and the case bounds), and evaluates the box minimum in exact rational arithmetic for each of the 60 cases. In two independent audits the check rejected mutated row tags and a case with the rows of Corollary 14 removed.

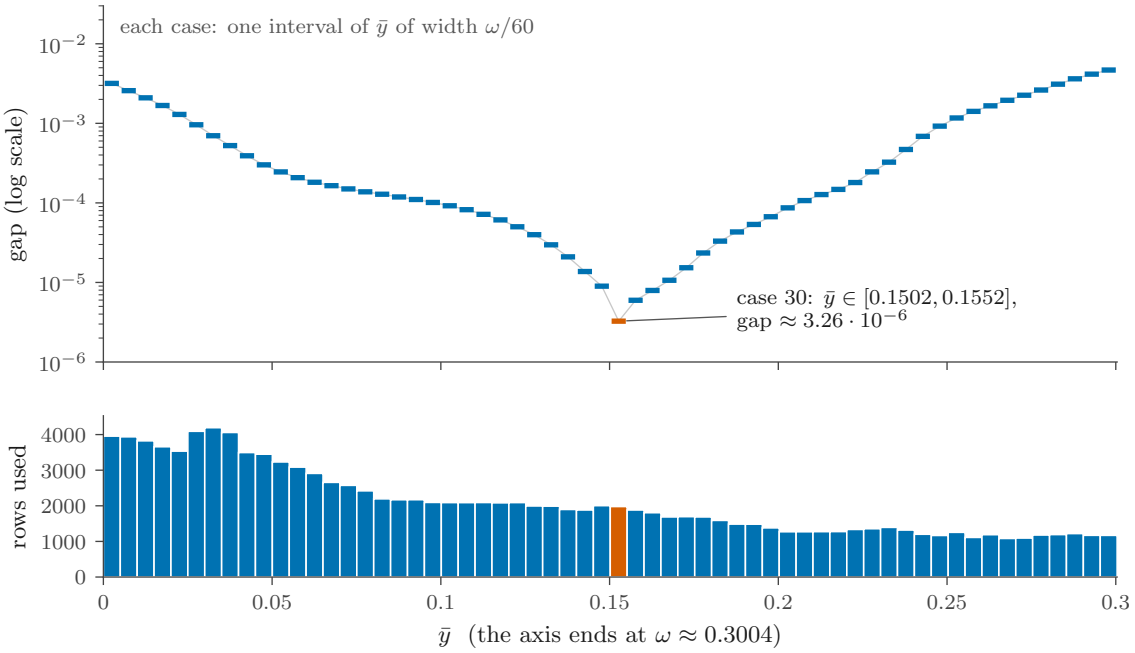


Figure 10: The 60 cases of the certificate. Top: the gap of each case (log scale), that is, the amount by which the weighted sum of the inequalities fails; every gap is positive. The middle cases, near $\bar{y} \approx 0.15$, are the tightest. Bottom: the number of inequalities that the certificate of each case combines.

Lean. The whole proof is formalized in Lean 4 with Mathlib. Each row family is a Lean lemma about an arbitrary configuration in general position under rational side conditions; a computable row builder maps a tag to a rational row, checks the side conditions, and is proved sound. For each of the 60 cases a single theorem checks the weighted combination by kernel reduction (`decide +kernel`), with the multipliers rounded to integers at scale 2^{200}. The final statement, `Esgk.numDistances_gt_one_third_add_kappa_image_gap`, says that for an injective configuration of $n \geq 10000$ points in general position, $(1/3 + 1/40)\, n < D$. Its axiom closure consists of the three standard axioms `propext`, `Classical.choice` and `Quot.sound`; the proof uses no `sorry` and no `native_decide`.

Availability. The certificate, the exact verifier and the Lean sources are at `https://github.com/mysticflounder/distinct-distances-general-position`.

10 Remarks

Limits of the method. With the same row families, the floating-point linear program has feasible points at $\delta = 1/39$ for $\bar{y} \in \{0.128, 0.154, 0.180, 0.205\}$. This indicates (floating point only; EMPIRICALLY VERIFIED in that sense) that these rows alone do not give $\delta = 1/39$ by this kind of certificate. Figure 11 shows the feasible point at $\bar{y} = 0.154$: it is one profile that these inequalities do not exclude at $\delta = 1/39$ (the one that the solver returned; others can exist). Its pair distances concentrate in three narrow ranges. At $\bar{y} = 0.154$, three further families of inequalities (Bochner positivity of the distance measure, a lower bound for short pairs from the area of the annulus, and a three-point covering inequality) each cut the feasible point found but leave the linear program

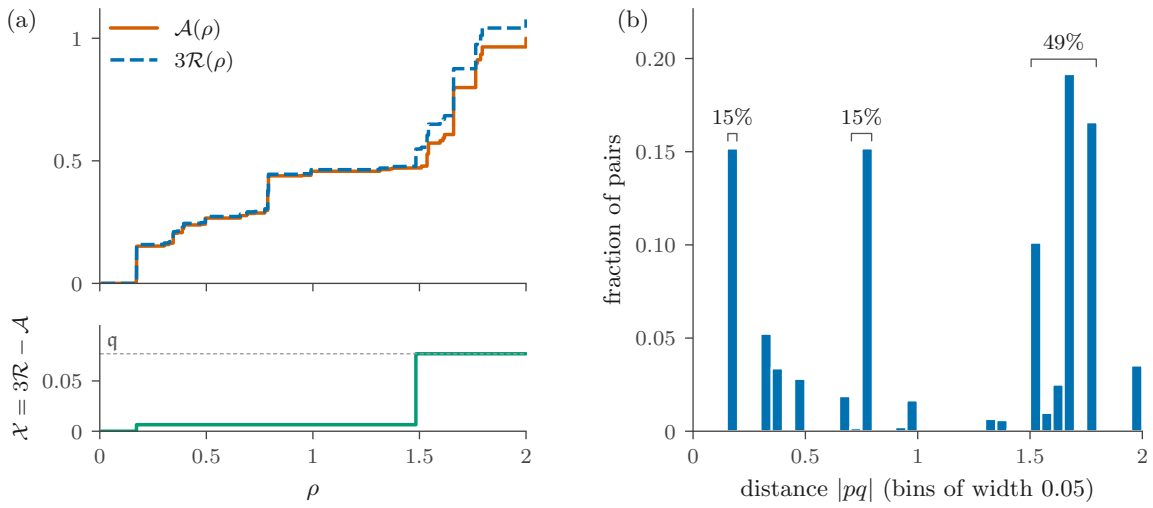


Figure 11: A solution of the linear system at $\delta = 1/39$ and $\bar{y} = 0.154$, found by a floating-point solver; in floating point, the inequalities of this paper do not exclude it. (a) The profile functions $\mathcal{A}$, $3\mathcal{R}$ and $\mathcal{X} = 3\mathcal{R} - \mathcal{A}$. (b) The distribution of the pair distances, as the increase of $\mathcal{A}$ over bins. It is not known to come from any point set.

feasible. HEURISTIC: a larger constant needs a further inequality of the kind of Lemma 13.

Pinned distances. Theorem 1 bounds the total number of distances. It gives no bound for the number $t(p)$ of distances from a single point, and the questions of Erdős and Pach whether $\max_p t(p) \geq (1/3 + \varepsilon)n$ or $\sum_p t(p) \geq (1/3 + \varepsilon)n^2$ for sets in general position [5, Problem 2]; I am not aware of an answer to either. The second question is tied to the number x_2 of pairs with two apexes. Let c_k be the number of pairs (p, r), $r > 0$, with $d_r(p) = k$. As in the proof of Lemma 2, $2x_0 + x_1 = \sum_{\{a,b\}}(2-\alpha) = \sum_{p,r}(d - \binom{d}{2})_{d=d_r(p)} = c_1 + c_2$, and $\sum_p(3t(p) - (n-1)) = \sum_{p,r:\, d_r(p)\geq 1}(3 - d_r(p)) = 2c_1 + c_2$. Since $x_0 + x_1 \leq 2x_0 + x_1 \leq 2(x_0 + x_1)$ and $c_1 + c_2 \leq 2c_1 + c_2 \leq 2(c_1 + c_2)$, the quantities $N - x_2$ and $\sum_p(3t(p) - (n-1))$ are within a factor 4 of each other. So a bound $x_2 \leq (1-c)N$ with $c > 0$ gives a positive answer to the second question, and a positive answer gives such a bound. Erdős and Purdy [3] remark that, for sets in general position, they cannot even show that the number of isosceles triangles is $o(n^2)$. The proof above uses the common set R of distances through (1) and does not bound x_2.

A Notation

Symbol	Meaning
P, n	the point set and its number of points
R, D	the set of distances of P and its size $D = \lvert R\rvert$
$C(p, r)$	the circle with centre p and radius r
$d_r(p)$	the number of points of P on $C(p, r)$; at most 3
$\mathrm{def}_r(p)$	$3 - d_r(p)$, the empty slots on $C(p, r)$
Q	the deficit $3D - n + 1$, the number of empty slots around each point
$\alpha(a, b)$	the number of apexes $z \in P$ with $\lvert za\rvert = \lvert zb\rvert$; at most 2
$\mathrm{bis}(a, b)$	the perpendicular bisector of a and b
N, x_k	the number $\binom{n}{2}$ of unordered pairs, and the number with $\alpha = k$
$\mathcal{D}$, κ	the pairs with two apexes, and the map to their pair of apexes
S	$\sum_p \lvert p\rvert^2$ after the centroid is moved to 0
m_p	the missing moment $n^{-1}\sum_r \mathrm{def}_r(p)\, r^2$; $\lvert p\rvert^2 = 1 - m_p$
$\bar{s}$, ω	the inner radius 41821/50000 of the annulus and $\omega = 1 - \bar{s}^2$
y_p, $\bar{y}$	the height $\lvert p\rvert^2 - \bar{s}^2 \in [0, \omega]$ and its average over P
$\mathfrak{d}$, ε, $\mathfrak{q}$	D/n, $1/n$ and Q/n
$\mathcal{A}(\rho)$	the number of ordered pairs of distinct points at distance at most ρ, divided by n^2
$\mathcal{R}(\rho)$	the number of distances at most ρ, divided by n
$\mathcal{X}(\rho)$	$3\mathcal{R}(\rho) - \mathcal{A}(\rho)$: the empty slots on circles of radius at most ρ, summed over all points, divided by n^2
$C(I, \tau)$	for every $p \in P$ and $r \in R \cap I$ with $d_r(p) = 3$, two of the three points on $C(p, r)$ are within τ
σ_0, Λ_0	the bounds for the distance between the two apexes in Lemma 13
G	the grid of 1283 radii used by the certificate

B Status of the ingredients

Obligation	Statement	Label	Evidence
O1	Lemma 2 (apex identity)	PROVEN (paper, Lean)	§2
O2	Lemma 3 (κ injective)	PROVEN (paper, Lean)	§2
O3	Lemmas 4, 5 (annulus)	PROVEN (paper, Lean)	§3
O4	Lemmas 6, 7 (chords)	PROVEN (paper, Lean)	§4
O5	Lemmas 8–11	PROVEN (paper, Lean)	§5
O6	Lemma 12 (apex bands)	PROVEN (paper, Lean)	§6
O7	Lemma 13, Corollary 14	PROVEN (paper, Lean)	§7
O8	the 60 case certificates	PROVEN (exact check, Lean kernel)	§8, §9
O9	Theorem 1	PROVEN (paper, Lean)	§9
O10	no certificate at $\delta = 1/39$ from these rows	EMPIRICALLY VERIFIED	floating-point LP only

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